

Workshop on Public Health Policy, Risk Perception, and Infectious Disease

Workshop Report

The Organizing Committee*

Essex Resort and Spa, Essex, Vermont, 10–12 August 2026

Executive summary

Infectious disease is a coupled human–pathogen system for which risk perception, social processes, and public policy shape transmission and are shaped by it. In most cases, though, the quantitative models that assist in public health response treat policy as an external switch and typically are constructed without ongoing involvement of public health professionals responsible for guidelines and interventions. We convened a two-day workshop with approximately thirty academic modelers, social scientists, and epidemiologists and public health professionals from federal and state agencies (the Centers for Disease Control and Prevention, the U.S. Geological Survey, and four state health departments) with the following objectives: (1) to foster sustained dialogue among academics, government epidemiologists, and non-governmental partners; (2) to ground mathematical modeling in a realistic understanding of how policy is formed and implemented; (3) to identify concrete ways in which modelers can better support epidemiologists and decision-makers outside academia; and (4) to consider models that deal with the interactions between policy, human risk perception and behavior.

Discussions at the workshop ranged broadly and this report categorizes eight overall results from these concerning policy matters (1–4) as well as models, data, and their impact on policy development and implementation (5–8). Our main **takeaways** are:

1. **Trust and relationships are a precondition for use.** Models and tools are used when they are co-created. Trust and relationships, formed through ongoing interactions and anchored in durable institutional levers (department-university agreements, internships and rotations, joint projects with a resulting product passed on to the public health agency), enhance the potential for a decision support tool to be used.
2. **“Policy” has multiple levels.** “Policy” is not a clearly defined single term. It can refer to an overall authority that sets major goals and constraints (sometimes referred to as big-P) or more tactical implementation guidance (sometimes referred to as little-p). The term also may refer to rules and interventions created by an agency to impact uptake and individual action, programs designed to affect social norms, and methods to provide information such as educational campaigns. Quantitative models have greater potential for impact if they acknowledge these multiple policy levels.
3. **Agency, effort, and scale differ across policies.** There are several dimensions which affect policy formation and impact. These dimensions are often generalized as a continuous gradient between top-down policies or structural changes (specified by an authority to force

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individual action) and bottom-up policies (arising due to people’s voluntary individual decision processes, such as after a public messaging campaign). This gradient can enforce false dichotomies about public health policy: centralized (federal) versus decentralized (local) policy, mandates versus recommendations, etc. This gradient can also conflate major policy considerations such as how an intervention affects individual agency (e.g. an individual’s capacity for determining action), changes in individual effort devoted to avoiding health risks, and the scale at which an intervention is designed to be implemented. For example, public health practitioners may well prefer policies that implement public messaging campaigns to reduce disease transmission because these interventions preserve the agency of individuals.

4. **Clarify policy objectives and trade-offs.** Minimizing infections may not be the only or even the main goal. Policy objectives are subjective, affected by the current social and political climate. Policy objectives may also misinterpret the wants and needs of constituents. Factors including acceptability, fatigue, workforce impacts, learning loss, capacity for individual action, and the decision-maker’s risk tolerance may all play a role in policy formation and impact. As a result, policy objectives are often opaque and decision-makers prefer relative assessments of the impacts of feasible options with ranges, not a single “optimal” answer. Clarifying policy objectives will help make trade-offs more explicit and may explain differences in public opinion about public health policies.
5. **Heterogeneity and uncertainty shape outcomes.** Population heterogeneity and uncertainty of behaviors across the population shape potential outcomes and evaluation of the impact of policies. Separating the impact of demographic heterogeneity (e.g. age, economic status, etc.) from the impact of stochasticity due to lack of knowledge of social processes is a challenge. Ideally, models should have the capacity to report the probability that a benefit materializes and consider population sub-groups (e.g. the never-compliant, movable-middle, and will-follow-through) as either dynamic variables that change as the policy is implemented, or as adjustable parameters to ascertain the impacts of heterogeneity on whole population response.
6. **Multiple models are good.** No single type of model dominates in analysis of policy impacts. Results indicate that in general ensemble averages across many models with differing structures and detailed assumptions are more predictive than any single model consistently can be. As with any modeling exercise, clear documentation in advance of model objectives, assumptions and criteria that evaluate the model utility for public health would benefit intermodel comparisons.
7. **Data are beneficial as a constraint on model generality.** Data should be considered as a major binding constraint on model capabilities. Surveillance and survey capacity across public health agencies may not be sufficient to provide clear differentiation between alternative model structures or assumptions. However, access to serology, health and claims records, school absences and business sick leave records, and routine monitoring or surveying of health behaviors or public opinion provide candidate means for infectious disease models to be aligned with observation. Public health practitioners have extensive qualitative knowledge and intuition that assist model formulation beyond explicit quantitative data.
8. **Communication is a key component in modeling.** Effective communication is best viewed as part of the modeling process, not an ancillary action taken after a model is developed and analyzed. Communicating model results and insights to decision-makers requires different approaches from communicating models to decision-makers. Not everyone speaks in the language of models or appreciates the underlying assumptions and constraints.

For example, it may not be effective to communicate model results to policymakers by focusing on specific model parameters or equations rather than a graphical analysis of potential impacts. Communication of the model itself may become important as a means of assessing the generality of insights that have been deemed valuable to policymakers. Suggestions for being effective in communicating model results and insights to decision-makers include utilizing trusted messengers who have a prior-developed rapport, providing ranges for model response, illustrating results with at most three comparisons, not using the term “model,” considering how different displays of uncertainty modify what people conclude, and designing the question you are addressing so that its implication fits in a single sentence.

1 Introduction

Purpose. Infectious disease spread arises from coupled human–pathogen systems in which risk perception, social processes, and policy shape transmission and are shaped by it in turn. The models that inform public health response have begun to represent human behavior, but they generally still treat policy as an externally-specified switch, and the people involved in setting and implementing policy are rarely directly involved in the model-development process. This workshop, convened under a National Science Foundation project on modeling human behavior and policy in infectious disease (Award 2436120), brought together about thirty academic modelers and social scientists, epidemiologists from federal and state agencies, and communication professionals to consider how policy is actually formed and implemented, how models are used (or not), and what would make models more useful to the public health professionals as they develop and implement policy.

Objectives.

1. Foster sustained dialogue among academics, government epidemiologists, and non-governmental partners.
2. Inform modeling efforts by including a realistic understanding of how policy is formed and implemented as a function of epidemiological and social conditions.
3. Identify concrete ways in which the modeling community can better support epidemiologists and decision-makers outside academia.
4. Consider how policy interacts with risk perception and the behavioral changes that ameliorate (or fail to ameliorate) disease spread.

Format. Three framing talks, three panels, and four breakout sessions filled two days (Appendix A). Each breakout time had two groups in parallel: the first breakout gave both groups the same question, the later three breakouts gave each group separate questions. Leadership team members took notes during each session and reported a summary of the discussion to the full workshop. Two artists from Event Rap performed a “rap-up” of each session, and a peer-reviewed rap video produced for the project’s broader-impacts effort premiered on the first morning. Sources and method are described in Appendix B.

Participants. *Organizers:* Brian Beckage, Ari Freedman, Laurent Hébert-Dufresne, and Greta Savitsky (University of Vermont); Louis Gross, Suzanne Lenhart, and Charles Sims (University of Tennessee, Knoxville); Katherine Lacasse (Rhode Island College); Chadi Saad-Roy (University of British Columbia, virtual). *Agencies:* Matthew Biggerstaff, Jamison Pike, Philip

Salvatore, and Rachel Slayton (Centers for Disease Control and Prevention); Richard Berl (U.S. Geological Survey); Emily Mosites, Jessica Schifano, and Katie Warchut (Vermont Department of Health); Tomás León and Lauren White (California Department of Public Health); Ken Komatsu (Arizona Department of Health Services); Catherine Brown (Massachusetts Department of Public Health). *Academia*: Sara Constantino (Stanford University); Gökçe Dayanıklı (University of Illinois Urbana-Champaign); Baltazar Espinoza (University of Virginia); Bryan Grenfell, Emily Howerton, and Simon Levin (Princeton University); Stephen Kissler (University of Colorado Boulder); Bouchra Nasri (University of Montreal); Elizabeth Talbot (Geisel School of Medicine at Dartmouth). *Communications*: Baba Brinkman and Dex McBean (Event Rap); Catherine Crawley (public information officer).

2 Framing talks

Beckage presented the project’s view of disease as a coupled human–pathogen system: a hierarchy of behavioral processes (information, cognition, contagion, trust, perceived risk, culture, and governance), a model of perceived risk driven by the frequency and severity of events, and an opinion-group SEIR model in which people move among groups that oppose, are undecided about, or favor protective behavior according to the balance of perceived disease risk against perceived risk of the behavior. Policy in these models is still exogenous; the meeting’s stated objective was to learn how to endogenize policy as the behavioral response of a collective intelligence. Sims discussed some connections to economics. A summary is that mental decisions may differ from behaviors (observed), policies (formal rules) from agencies and social norms, and two feedback loops, between risk assessment and risk management and between individual risk management and public policy, are what ideally models should capture. Policy is “clunky” (turned on, left on, turned off), so epidemic curves may be chopped rather than flattened, and the motivation for policy is to act in situations for which private risk management falls short. Lacasse reported a discrete-choice experiment on mask wearing. Key results on participant responses about factors that most impact intent to wear a mask were that mortality rate mattered most, then the presence of children, infection rate, a mandate, and the social norm, with respondents falling into disease-risk, protect-the-vulnerable, and read-the-room profiles. The following discussion noted the need for careful vocabulary, particularly the distinction between “big-P” policy (legal authority) and “little-p” guidance implemented by partners under their own constraints and the practice of several agencies to avoid using the word “policy” in favor of “intervention” and “recommendation.”

3 Breakout sessions

3.1 Breakout 1: How to enhance interaction among policymakers, epidemiologists, and modelers to co-create decision support tools (both groups)

Both groups asked how policymakers, epidemiologists, and modelers can co-create decision support tools that are actually used, and both reached similar answers: most such tools are never used, and the ones that are were built from the start with the decision-maker, the decision context, and the feasible interventions defined. The USGS practice of a value-of-information analysis before any modeling was offered as a template. Trust and relationships are the precondition and are built over time; because leadership turns over, relationships with durable staff matter more than those with a champion at the top of the hierarchy. Institutional levers were suggested as contributing to the co-creation and trust-building enterprise: “academic health department”

agreements between a state and a university (e.g. covering workforce training, staff teaching, and honest-broker data sharing), internships and placements in agencies, practitioners teaching university courses, and joint projects in which the academic partner builds a tool, hands it over, and teaches the department to use it.

Several cautions were mentioned in this process. Tools have to work and their failure modes have to be known. Examples included sepsis-prediction tools with high false-positive rates which were discarded wholesale, and discordant COVID models led at least one governor to refuse to hear about models at all. Public health policy is distinctive in its speed and in the unpredictable scope of expertise it needs (ventilation engineers, for instance), and adaptive-management cycles with thresholds for change were proposed as antidotes to “policy fatigue.” Whether trust is better served by one vetted modeling group (e.g. WHO-endorsed tuberculosis modelers) or by an ensemble (the IPCC model) is an open issue; ensembles show whether a ranking of interventions is robust, but their uncertainty must be digested for a decision-maker within short time periods. Objectives are subjective: the decision-maker’s values weight the trade-offs (infections as opposed to mask exhaustion and workforce preservation in hospitals), so trade-offs are best incorporated at the outset, and models should report outcomes for individuals and institutions as well as populations and respect the capacity of the department that must act. Simple tools often suffice (a state-level nowcasting tool proved almost as good as its academic counterpart), while a complex model can still give a clear answer (encampment clearance and overdose deaths). Publishing applied models is not easy because journals reward novelty. On public trust, both groups asked what public health can learn from meteorologists, who are wrong often and trusted anyway, and noted two differences: public health has coercive power, and its successes leave no visible counterfactual.

3.2 Breakout 2: How can models analyze uncertainty and heterogeneity in risk perception and behavioral response in policy frameworks? (Group 1) How do we use social-science, economic, and behavioral theory to build models for policy? (Group 2)

Group 1 asked how models can analyze uncertainty and heterogeneity in risk perception and behavioral response. Modelers should separate reducible uncertainty from irreducible stochasticity and report the probability that an intervention’s benefit materializes, not only its expected value, because one village can have ten times the cases of another under identical policies. Risk perception can be built from survey distributions or from psychological theory collapsed into an information-processing framework, and mean-field games can give follower, indifferent, and free-riding types objectives stratified by politics and income. What most models lack is waning and rebounding perceived risk, and what practitioners want is simpler: the uptake threshold below which an intervention fails. Little-p guidance captures only the willing while big-P mandates reach the skeptical but provoke reaction; little p becomes de facto big P when insurers or immunization law adopt a recommendation (the CDC opioid guideline). Guidance trickles down unevenly across jurisdictions and information provision itself, such as a public prevalence map, is a policy that acts on perceived risk. What turns a policy on or off is prevalence, the calendar, political pressure, or personality; predefined triggers (hospital-capacity thresholds, a blueprint at state-level to respond, CDC Community Levels) depend on receptive leadership. Suggested norms for modelers who partner with agencies included: keep the agency informed, follow clearance processes, and release nothing without an early conversation.

Group 2 took vaccine uptake as its case for bringing social, economic, and behavioral theory

into models. Political polarization and trust were the theories most mentioned as being needed, alongside the distinction between models that describe behavior mechanistically and models that ingest behavioral data. The populations for which data are missing may behave differently from those for which data exist; insular communities (the 2019 measles outbreak in upstate New York) show that opposition may arise from belief, targeted misinformation, or disconnection from care; and public health officers hold knowledge (e.g. summer camps with very limited nursing staff) that no dataset contains. The group argued for mixed methods and, provocatively, for making quantitative models subservient to qualitative knowledge. Socioeconomic barriers, institutions that turn recommendations into requirements, and the opportunity costs of shrinking departments all shape uptake, and a marketing analogy (accept that some demographic groups cannot be changed so focus on the persuadable middle) drew both interest and discomfort.

3.3 Breakout 3: Policy for integrative (One Health) systems that include human values, risk perceptions, and behaviors as both influences and outcomes of disease. (Group 1) Roles and coupling of process-based versus black-box approaches for modeling and policy. (Group 2)

Group 1 took tick-borne disease as its One Health case: ecologically complex, rising for a decade in the Northeast despite increased messaging, and apparently insensitive to individual behavior. The prevention framework works when used, but compliance is low (smell, hassle, fear of chemicals), and public health training rightly discourages reliance on sustained individual behavior change; ecological levers (deer reduction, predator restoration) have limited evidence of success, non-target effects, and their own “communication landmines.” Every state has a tick plan; the modeling question is how effective it is at realistic uptake and whether any of it is futile, and some states ask explicitly for tick-borne disease modeling to help prioritize a shrinking budget. New World screwworm, re-emerging after eradication, supplied the lesson that success disincentivizes the intervention that produced it, a parallel to childhood vaccination. The group distinguished one-time, daily, and seasonal protective behaviors, each with different motivators (a scare works to enhance a one-time action, but not for forming a habit).

Group 2 began from a rule of thumb: system-agnostic time-series models usually win for short-term forecasting accuracy, and mechanistic models win for ability to provide explanation. Black boxes are acceptable for short-horizon logistics (beds, staff, supplies) but not for counterfactual or comparative questions. Overfitting and distribution shift (a post-pandemic RSV season) are their main risks, and an AI-built model can fit the data well with impossible flows unless expert constraints are imposed. To decision-makers even a simple SIR model is a black box, so “who is the black box?” should be considered first. Evaluation should be tied to the objective and, where possible, to public health utility, and models of every type should document objectives and evaluation criteria, as agent-based models do under the ODD protocol. A live demonstration from the COVID-19 Scenario Modeling Hub was decisive: no one could tell the semi-mechanistic model from the one an undergraduate built with an AI assistant; the best model is never the same from round to round while the ensemble is consistently among the best; and scenario rankings are usually robust even when magnitudes differ. Hybrid and semi-mechanistic (“grey-box”) models are an appealing option, but science is needed to provide evidence of which black-box components are reasonable.

3.4 Breakout 4: How to communicate benefits, limitations, and policy implications of models to public health professionals and the public. (Group 1) The relative effectiveness of bottom-up versus top-down policies. (Group 2)

Group 1 produced practical guidance for communicating models: present results first and hold limitations for questions; never give a single number, give a range or scenarios that bound the outcomes; compare at most three options; and know the decision, the decision-maker, and their competing pressures before presenting. Reach policymakers through trusted public health professionals, who fold the model into a little-p recommendation and may never use the word “model”; hold short trainings (two hours at most) with a toy model before a crisis arises. Good communication starts when the question is chosen: the CDC’s Morbidity and Mortality Weekly Report (MMWR) protocol of stating what is known, what is added, and what the reader should do applies to models too. Research needs suggested as benefiting from “a pot of money” were communication science on how uncertainty displays change what people conclude, better ways than dollars to quantify trade-offs (learning loss as compared to infections; quarantine duration as compared to risk tolerance), and rotations of modelers into agencies and agency staff into universities.

Group 2 reframed the issue: the top-down versus bottom-up dichotomy comes from modelers. A more useful axis is how much agency a policy gives people, with fluoridation (no agency of individuals, no effort of individuals) and a mask mandate (high individual agency as well as high effort) as endpoints on this axis and with effort and the scale of governance as possible further dimensions for the space of policies. Several cases discussed blurred the dichotomy of top-down and bottom-up. Arizona’s tribal data-sovereignty policy is a top-down structure that gives tribes bottom-up ownership of their data. Whether individual agency reduces divisiveness is not clear. Even optional chronic-wasting-disease testing provokes furious opposition, and mandates breed pockets of resistance whose blowback (rescinded military vaccination requirements) can set policy back. Agency can also amplify inequality. An example was that under stay-at-home guidance the essential workers of one Vermont town moved more and were blamed for it, and people experiencing homelessness (a small fraction of the overall population but sometimes having a large share of harmful bacterial disease) have almost no agency to act on any recommendation. While public health institutions may prefer systems-level changes that do not restrict individual agency (ventilation, repellent dispensers at building exits), changes to material culture can motivate changes through a nudge. Evidence-based decision-making may favor uniform top-down interventions because they are easier to study, while bottom-up heterogeneity is where the statistical power lies. The group closed with a challenge to modelers: our models wash individual agency away in the mathematics and are agnostic between “cannot” and “will not,” and we owe it to potential users of the models to be transparent about that.

4 Panel discussions

4.1 Panel 1: Panel of state epidemiologists, federal and non-governmental researchers—how is policy implemented in your state/agency? How does your state/agency consider, develop and use models in support of policy?

Ken Komatsu, Matthew Biggerstaff, and Catherine Brown, moderated by Louis Gross, described policy from the perspectives of their agencies. CDC’s influenza modeling informs little-p decisions including: the trade-off between a more effective vaccine that arrives later and convenient vaccination now, translating surveillance into burden so that risk is legible to the

public, and placing scarce testing to maximize detection. Massachusetts practices little p as risk communication, “advice from a friend” that encourages turning a behavior into a routine, and reserves big P for cases such as outdated rabies regulations, where a rule change “flipped a switch” to modify overall response. Arizona’s statutes confer broad authority, including the power to restrict certain individual rights, and generate the surveillance data on which everything rests. The panelists noted differences in defining policy and the CDC generally avoids the word because it can trigger very formal and time-consuming review. What makes a tool usable is based on politics and personality: find a champion, explain the limitations, never say “optimal,” prefer the term “evaluation” to “validation,” and present relative assessments of feasible interventions rather than absolute forecasts. During COVID, academics who “came out of the woodwork” with models could not be trusted because they were unknown. A hybrid of in-house staff and known academic partners, backed by ensembles of models with differing assumptions and limitations, was the preferred structure. Examples of modelers answering the wrong question (how many contact tracers, with a fourteen-day test delay; where to site vaccine trials) led to the panel’s central request: ask us what our questions are, and teach us what models can answer. Vermont’s citizen legislature has no capacity to absorb sophisticated models and wants near-binary answers, and there can be an “alignment problem” between academic goals and practitioner needs.

4.2 Panel 2: Panel on alternative means to incorporate beliefs, access to (mis)information, and behaviors in useful models for policy decision-makers

Jamison Pike, Simon Levin, Rachel Slayton, Gökçe Dayanıklı, Richard Berl, and Baltazar Espinoza, moderated by Katherine Lacasse, noted six approaches: embedding behavioral and economic instruments in surveillance networks; multi-level models of overlapping groups whose norms coevolve with polarization; simple, measurable behaviors that explain core results (healthcare workers moving among facilities); individuals as decision-makers with their own objectives in mean-field games; twenty-five years of hunter license and harvest data linked to regulation; and the triggers and daily-flipping reactions that make behavior a multi-scale problem. Payment to individuals for vaccination can sometimes be an incentive and sometimes a punishment, invites attributions of malintent, and may crowd out other motives. Interventions can change the norms they act on, so a standing program of social monitoring was proposed to follow this dynamic. Practitioners argued for data before models when uncertainty is high, for community leaders over individuals, and for a focus on the “movable middle” whose beliefs are not strongly held. Whether a question to be addressed requires inclusion in models of explicit behaviors led to differences of opinion. Scenario comparisons at an assumed level of vaccine coverage may not require explicit inclusion of behaviors while tracking an epidemic over time often benefits from including behavioral changes, and behavior is an implicit prior in every model regardless. One proposal is to have models with “sliders” for the compliant fraction and a search for the thresholds at which an incentive’s effect flips.

4.3 Panel 3: Most interesting takeaways from the workshop, what did we learn?

Stephen Kissler, Bryan Grenfell, Emily Howerton, Philip Salvatore, and Bouchra Nasri, moderated by Louis Gross, offered takeaways. Be practical, through relationships built over time so the trust is in place when a crisis needs to be addressed. Susceptible depletion gives epidemics a fundamental predictability that behavior cannot override, but we are in an era in which obtaining certain public health data is becoming challenging, and we can benefit from the use

of serology data. Becoming the meteorologists of public health may be a goal: prediction that is not always right but more reliable than not. For weather prediction the use of technology such as satellites brought higher predictive capability and a “satellite” for infectious disease would be methods to include individual behaviors and how these impact and are impacted by social norms. Data systems to be encouraged and reinforced include electronic health and claims records, which are often an untapped resource. The presence of practitioners changed academic participants’ sense of which questions to ask, and a similar workshop was encouraged to be held, with students present. Discussion considered whether modelers should model their own interactions with government, and whether a more effective lever is the environment rather than behaviors (e.g. Paxlovid uptake was limited less by fear of rebound than by a large out-of-pocket cost for some without insurance coverage). Public health might be viewed not as a primarily medical discipline but as an interdisciplinary enterprise including behavioral science, economics, and engineering. The group noted that communication can be considered an intervention with a causal pathway and only a small fraction of models ever reaches deployment.

5 Commitments and next steps

The organizers committed to composing this report; developing a meeting announcement on EurekAlert! (this was published in August 2026); a participant survey and summary of this to be reported to NSF; and wrap-up rap videos to be reviewed by participants before release. A perspective piece was agreed in principle, with venue and lead author to be settled once the messages are fixed. Candidate messages for this piece include trust as a long-term commitment, the importance of individual agency and the fracturing of opinion groups, unmet modeling needs, and a reframing from changing behavior to understanding it in order to improve welfare. Proposals from the floor for further activities included a short piece encouraging applied-mathematics students to intern in public health agencies, a routine list of open questions from state and local departments, meetings attached to professional-society conferences, a follow-on meeting with an international dimension, and a recurring workshop that brings agency staff and students together.

Acknowledgments

We used artificial intelligence (AI) tools at two stages in preparing this report. First, Plaud captured and converted the audio of the sessions into text transcripts. Second, we used an AI assistant (Claude, Anthropic) to synthesize these AI-generated transcripts with the human-written notes taken by members of the organizing team and the public information officer, and to draft and edit the report. We have reviewed and edited the output and take full responsibility for the content of this report.

A Agenda

Monday 10 August. Project leadership meeting; welcome reception.

Tuesday 11 August: framing, listening, and cross-sector dialogue

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| 9:00–10:15 | Welcome and goals (Beckage, Freedman). Framing talks: Beckage, a framework for human behavior in models; Sims, policy, economics, and infectious disease; Lacasse, experimental data on responses to infectious disease |
| 10:30–12:00 | Panel 1: Panel of state epidemiologists, federal and non-governmental researchers—how is policy implemented in your state/agency? How does your state/agency consider, develop and use models in support of policy? Komatsu, Biggerstaff, Brown; moderator Gross; rap-up |
| 13:30–15:00 | Breakout 1 (two groups, same question): How to best enhance interaction between policymakers, epidemiologists, and modelers to co-create decision support tools towards effective policy? Group reports and rap-up |
| 15:15–17:00 | Breakout 2. Group 1: How can epidemiological models enhance analysis of the impacts of uncertainty and heterogeneity in risk perception and behavioral responses in policy frameworks? Group 2: How do we more effectively utilize social science/economic/behavioral theories and findings to enhance building epidemiology models in support of policy? Group reports and rap-up; wrap-up |
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Wednesday 12 August: building, synthesizing, committing

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| 9:00–10:15 | Panel 2: Panel on alternative means to incorporate beliefs, access to (mis)information, behaviors in useful models for policy decision-makers. Levin, Espinoza, Dayanikli, Berl, Slayton, Pike; moderator Lacasse; rap-up |
| 10:30–12:00 | Breakout 3. Group 1: How can we more effectively incorporate and develop policy to support integrative systems of infectious disease and health (e.g. the One Health approach to humans, wildlife, livestock, land use, conservation) that include human values, risk perceptions, and behaviors as both influences and outcomes of disease processes? Group 2: What are the roles and potential coupling of process-based (mechanistic) vs. black-box (machine-learning/AI) approaches for epidemiological modeling and policy decision-making? Group reports and rap-up |
| 13:30–15:00 | Breakout 4. Group 1: How to communicate benefits, limitations, and policy implications of epidemiological models to public health professionals and the general public? Group 2: How might we consider the relative effectiveness of bottom-up vs. top-down policies (facilitating people to take action vs. mandating people take actions or structural changes)? Group reports |
| 15:30–17:00 | Panel 3: Most interesting takeaways from the workshop, what did we learn? Kissler, Howerton, Salvatore, Grenfell, Nasri; moderator Gross. Commitments and next steps; final rap-up (Brinkman, McBean) |
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Thursday 13 August. Departures; organizing-team debrief.

B Sources and method

Every panel and breakout was recorded on one of two audio recorders (one per breakout room) and transcribed; note-takers from the organizing team kept written notes for each session, and the public information officer kept notes across both days. Three plenary slots (the day-

one report-outs and wrap-up, the day-two Breakout 3 report-outs, and the day-two opening announcements) were not audio recorded but written notes were taken. Transcripts were cleaned and a document of substantive points was compiled for each session. The organizers used an AI assistant (Claude, Anthropic) to help clean the transcripts and compile the session documents into an initial draft, checked the content against the recordings and notes, revised the document and distributed it to the invited participants. Session documents and transcripts are available to participants on request.